

Lec 1

Aug 15, 2026

Books

1. Nelson *

2. Kartim & Johnson Probs

3. Kohavi Async.

4. Morris

Adnan Sir.

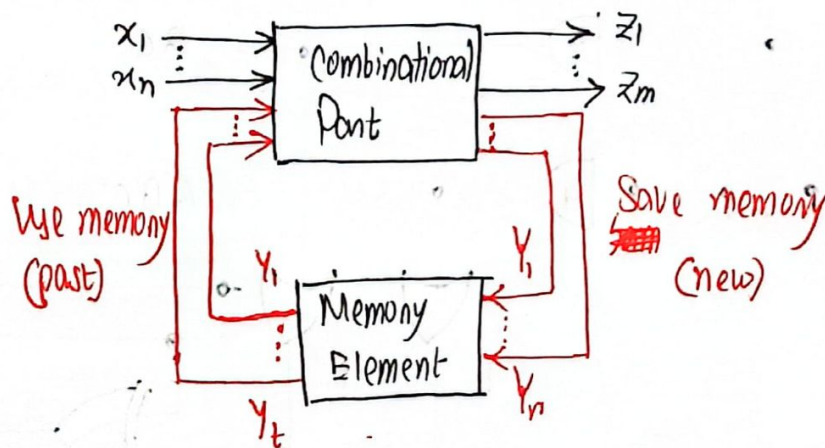
Sequential Circuit

Intro:

• Combinational $\rightarrow$ Output = $f(\text{Input})$

• Sequential $\rightarrow$ Output = $f(\text{Input, current state of the logic})$

$\rightarrow$ we need to save the current state in memory



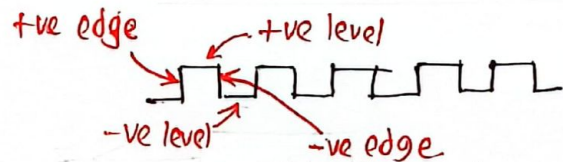
Sequential Circuit

Classification

- 2 Types $\rightarrow$ (Clock is used) Synchronous sequential circuit $\rightarrow$ (All memory elements change state at the same time)
- $\rightarrow$ Asynchronous sequential circuit (clock is not used)

Clock:

• Sequence of pulses



• Triggers $\rightarrow$ Level trigger $\rightarrow$ +ve, -ve
 $\rightarrow$ Edge trigger $\rightarrow$ +ve $\uparrow$, -ve $\downarrow$

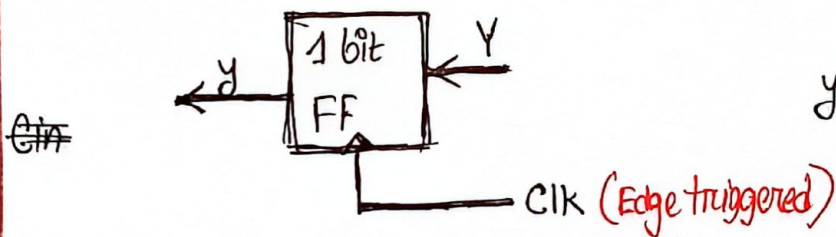
• To synchronize operations

Flip-flop:

• A memory element with ~~clock~~.

① Edge-triggered clock

~~latch?~~



Latch:

• Memory element with ~~no clock~~

① No clock

② Level-triggered clock



Types:

1. D Flip Flop : ~~y~~ $y = Y$

$\rightarrow$ Delay

2. SR Flip-Flop

$\rightarrow$ Set-Reset

3. JK Flip-Flop

4. T Flip-Flop

Types:

1. D Latch

2. SR Latch

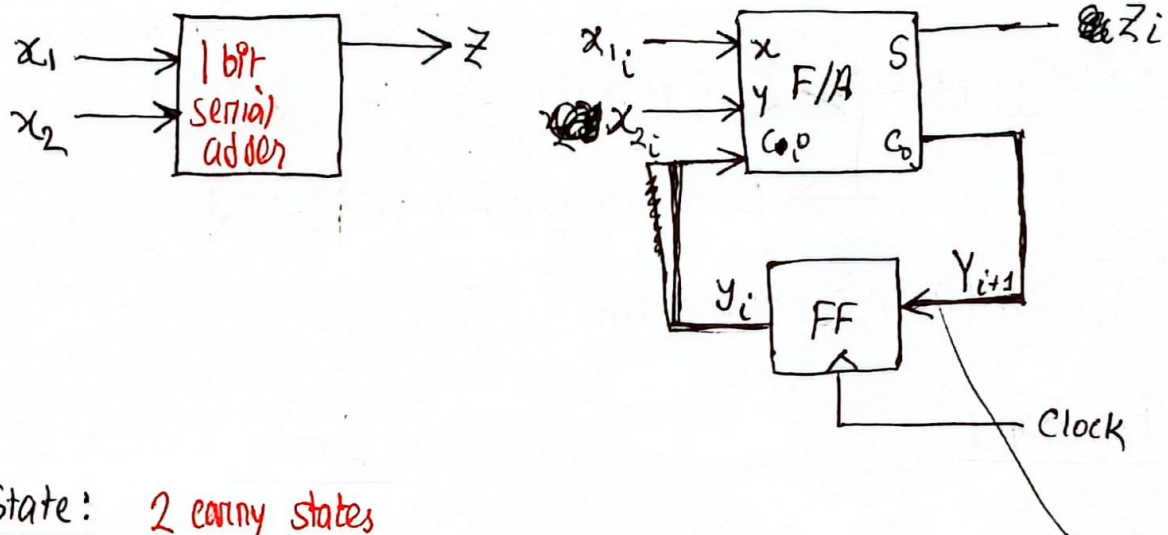
3. JK Latch

4. T Latch.

$\left. \begin{array}{l} \text{3. JK Latch} \\ \text{4. T Latch} \end{array} \right\} \text{Rarely used}$

More details later.

Example: 1 Bit Sequential Adder: Adds 2 binary numbers 1 bit at a time.

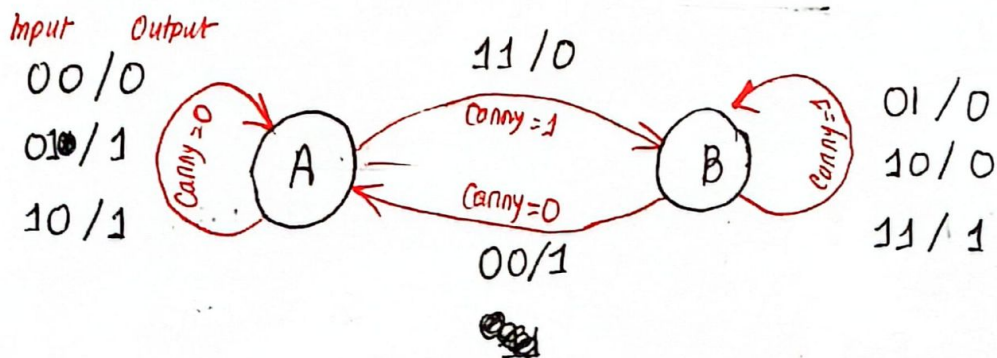


State: 2 carry states

"A": if carry = 0 at t_{i-1}

"B": if carry = 1 at t_{i-1}

State Diagram:



State Table:

Present state

x_1, x_2		NS, Z			
		00	01	11	10
$y=0, A$	$y=0, A$	A, 0	A, 1	B, 0	A, 1
	$y=1, B$	A, 1	B, 0	B, 1	B, 0

We could do vice versa as well.

Excitation Table:

$$\#FF = \lceil \log_2 n \rceil$$

Flip-flops needed $\rightarrow$ #states

Here, $n=2$ $\therefore$ #ff = 1

After K-Map:

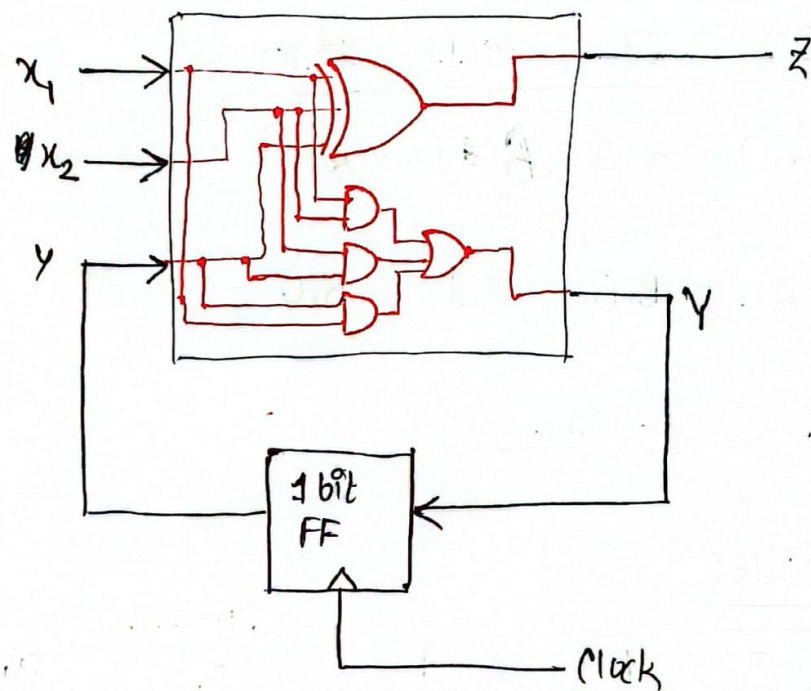
$$Y = x_1 x_2 + x_1 y + x_2 y$$

$$F = \dots = x_1 \oplus x_2 \oplus y$$

x_1, x_2		Y (next state)				Z (output)			
		00	01	11	10	00	01	11	10
y	0	0 A	0 A	1 B	0 A	0	1	0	1
	1	0 A	1 B	1 B	1 B	1	0	1	0

Carry Sum

Circuit Diagram:

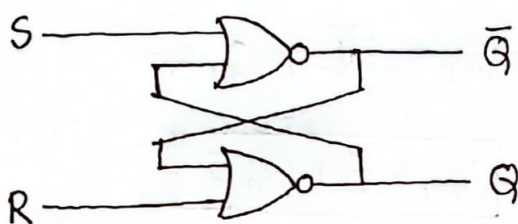


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Flip-flop and Latch

Latch
~~SR Flip-Flop~~

→ Set-Reset



→ it's the complement of Q for all cases except the Don't care / Not allowed state

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

NOR → output 1 only when both inputs are 0

Excitation Table:

Excitation input		Present state		Next state	
S	R	Q_n	$\bar{Q}_n$	Q_{n+1}	$\bar{Q}_{n+1}$
0	0	0	1	0	1
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	1	X	X
1	1	1	0	X	X

No change / Hold

Reset

Set

Not allowed

$R = \bar{Q} = 0$ keeps $Q = 1$

→ If $Q = 1 \rightarrow \bar{Q}$ is forced to be 0
If $Q = 0 \rightarrow S = \bar{Q} = 0$ makes Q to be 1
 $\bar{Q} = 1$ keeps $Q = 0$

→ $R = 1$ forces Q to be 0
 $S = \bar{Q} = 0$ makes $\bar{Q} = 1$

→ $S = 1$ forces Q to be 0
 $R = \bar{Q} = 0$ makes $Q = 1$

→ S, R both being 1 forces both NOR to output 0, this is perfectly stable. However, making both 0 suddenly causes race condition.

Both NOR has two 0 inputs, thus it'll try to output 1, which is feedbacked to the other NOR, which will be forced to output 0.

So, the NOR that responds first will win. Due to this unpredictable state

Race condition after switching $S=R=1$ to $S=R=0$, $S=R=1$ is ~~not~~ not allowed.

Characteristics Equation

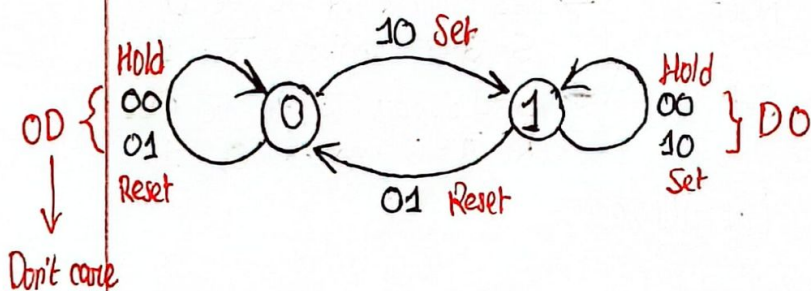
SR				
	00	01	11	10
Q_n	0	0	X	1
$\bar{Q}_n$	1	0	X	1
	R		S	

$$Q_{n+1} = S + \bar{Q}_n R$$

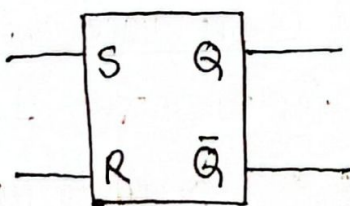
$$Q_{n+1} = S + Q_n \bar{R}$$

↑ No change ↑ Reset ↑ ~~Not~~ Not allowed ↑ Set

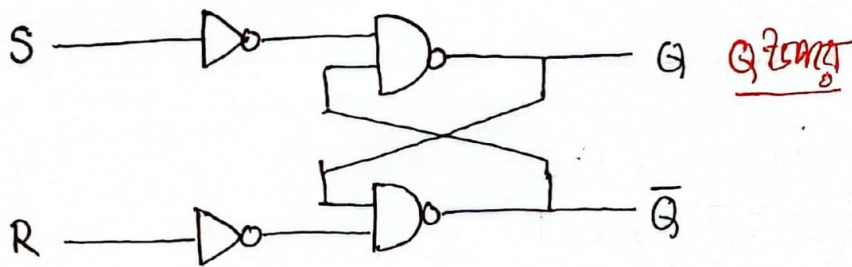
State Diagram: 2 states of Q



Symbols



SR Latch Using NAND Gates



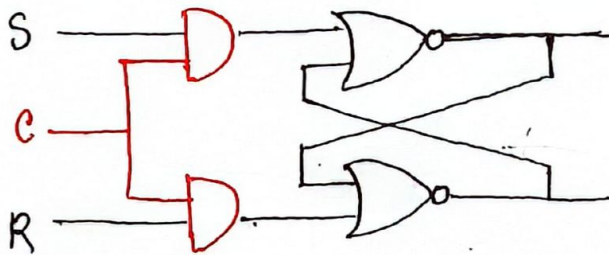
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NAND → (सत्यता) (वर्ग)
0 शक्ति 1

Same excitation table as the NOR one.

gated SR Latch

Controlled (with enabler / level triggered clock)



$C=0 \rightarrow$ AND's output = 0
No change according to normal latch

$C=1 \rightarrow$ AND's outputs = S, R
Behaves as normal latch

Excitation Table

Forcefully turned off

Normal Latch

C	S	R	Q_n	Q_{n+1}
0	X	X	0	0
0	X	X	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	X
1	1	1	1	X

Hold $\rightarrow$ because we kept off the latch, since $C=0$ will force AND's outputs = 0

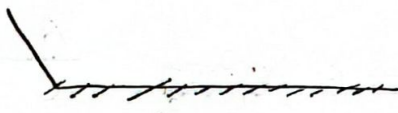
No change

Reset

Set

Invalid

◦ Characteristics Equation:

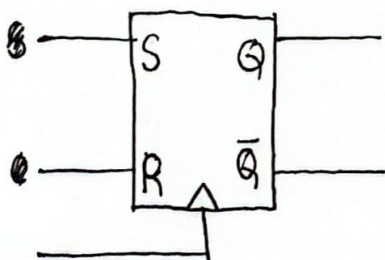

$$Q_{n+1} = \begin{cases} Q_n & , C=0 \\ S + Q_n \bar{R} & , C=1 \end{cases}$$

$$\begin{aligned} \therefore Q_{n+1} &= \bar{C} Q_n + C(S + Q_n \bar{R}) \\ &= Q_n (\bar{C} + C\bar{R}) + CS \\ &= Q_n (C + \bar{C})(\bar{C} + \bar{R}) + SC \end{aligned}$$

$$Q_{n+1} = SC + \bar{R}\bar{C} Q_n$$

◦ State Diagram:

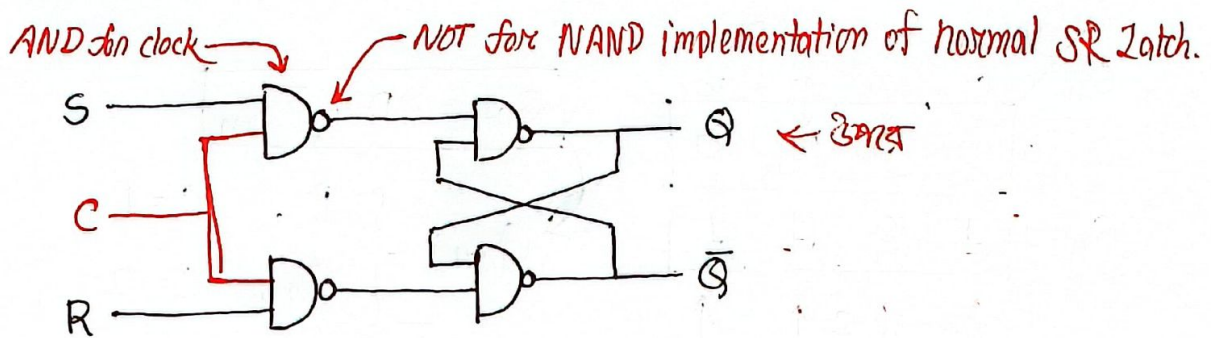
◦ Symbol: IC-74279



Dec 3

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◦ Gated SR Latch using NAND Gates:



D Latch: (Gated)

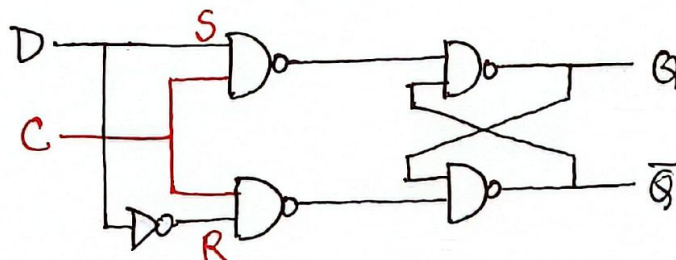
↳ Data.

$D=1 \rightarrow S=1, R=0$

$D=0 \rightarrow S=0, R=1$

∴ $S=D$

$R=\bar{D}$



◦ Excitation Table:

C	D	Q_n	Q_{n+1}
0	0	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Hold

Store 0

Store 1

Basically, we Store D by giving a clock pulse.

◦ Characteristics Equation:

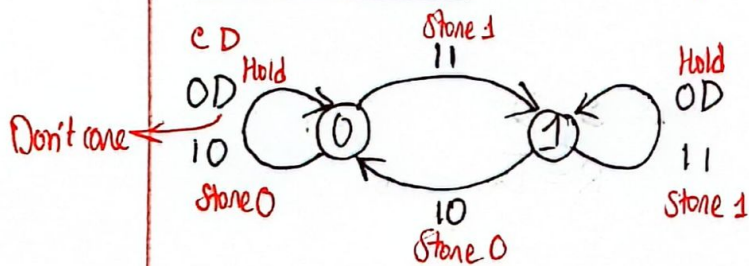
DC \ D	00	01	11	10
Q _n 0	0	0	1	0
Q _n 1	1	0	1	1

↑ ↑ ↑ ↑
 C=0 C=1 C=1 C=0
 ∴ Hold ∴ Store D ∴ Hold

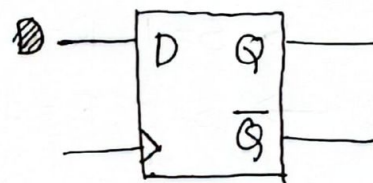
$$Q_{n+1} = CD + \bar{C}Q_n$$

$$Q_{n+1} = \begin{cases} D, & C=1 \\ Q_n, & C=0 \end{cases}$$

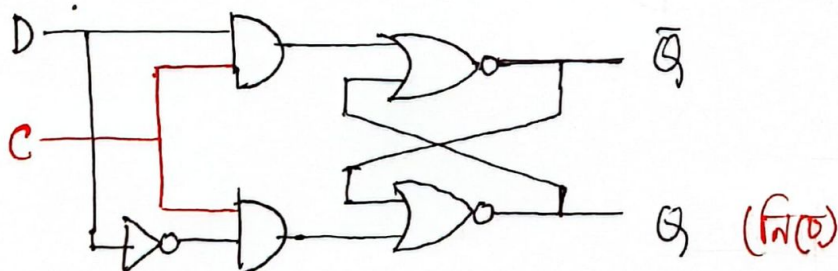
◦ State Diagram: 2 states of Q



◦ Symbol:



HW → ◦ D Latch using NOR Gates:

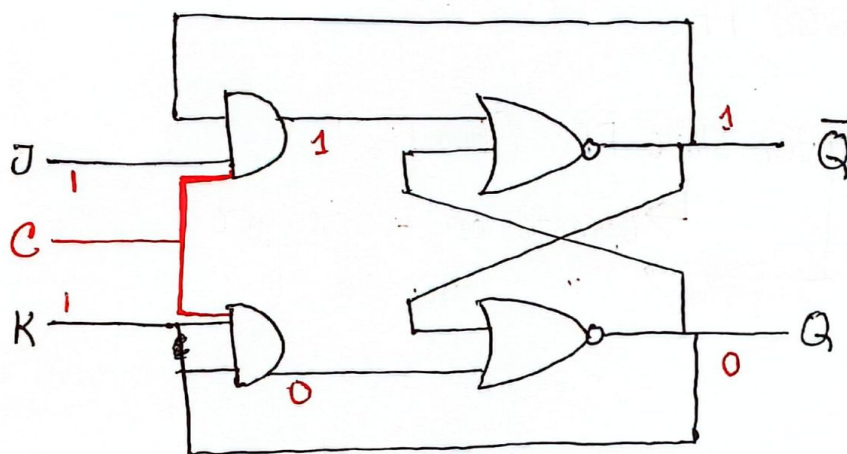


Lec 3

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JK Flip-Flop:

- Refinement of S-R FF
- Fixes the not allowed problem.



Excitation Table:

J	K	Q_n	Q_{n+1}	
0	0	0	0	No change
0	0	1	1	
0	1	0	0	Reset
0	1	1	0	
1	0	0	1	Set
1	0	1	1	
→ 1	1	0	1	"Toggle" ⇒ This causes alternating infinitely (continuous toggling)
1	1	1	0	

o Infinite toggle solutions:

• Add ~~to~~ clock

① Duration of level trigger clock should be ~~reduced~~ $<$ gate delay.

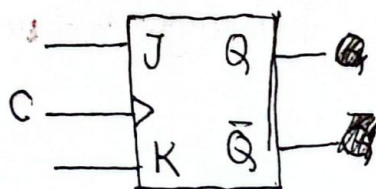
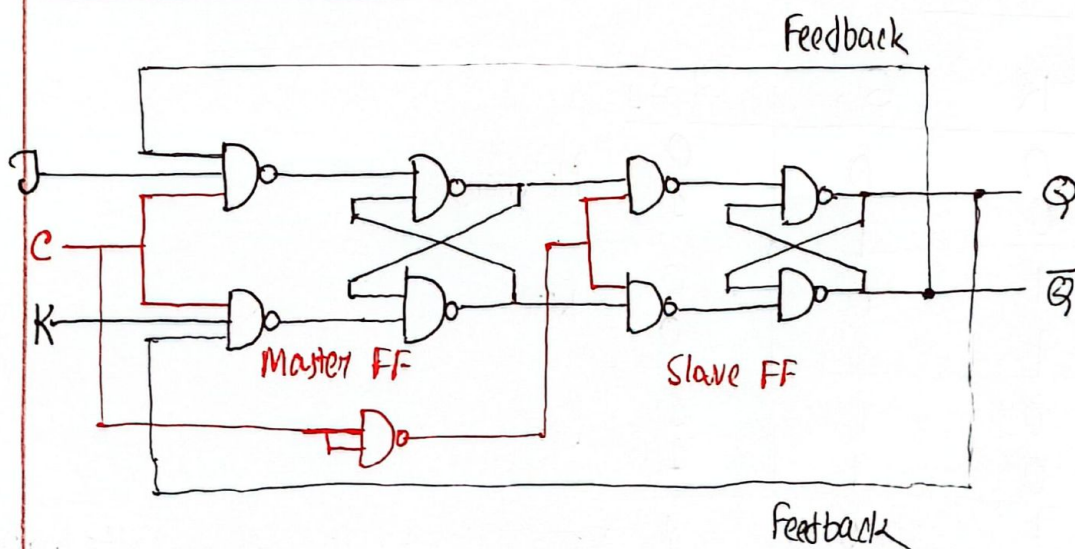
?

② Use Edge-triggered FF

③ ~~Master~~ Use Master-Slave FF (two FFs)

↳ Operates on -ve trigger
" " +ve "

Master-Slave JK FF:



Block Diagram

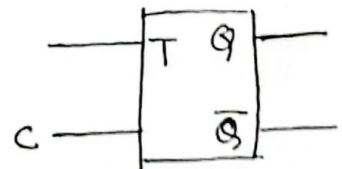
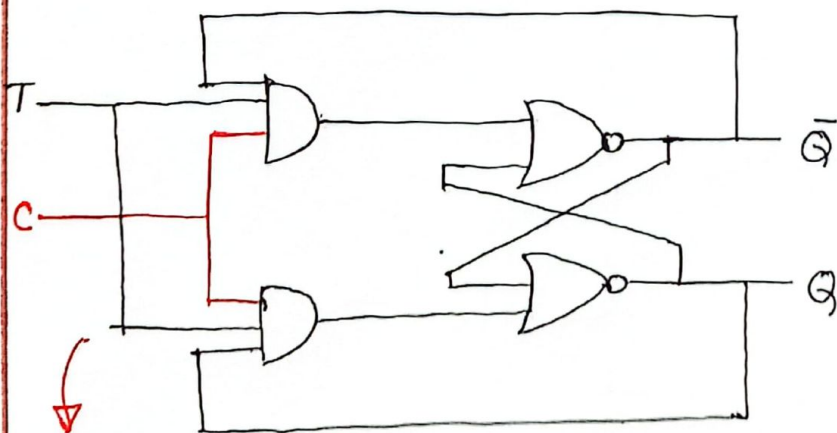
o Characteristics Equation :

JK Q _n				
	00	01	11	10
00			1	1
1	1			1

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$$

T Flip-Flop :

- Toggle Flip-Flop → When we do not need Set, reset feature
→ we only need toggle feature.



JK is shorted.

◦ Excitation Table:

C	T	Q_n	Q_{n+1}	
0	X	0	0	Hold
0	X	1	1	
1	0	0	0	No input change
1	0	1	1	
1	1	0	1	Toggle
1	1	1	0	

◦ Characteristics Equation:

$Q_n \backslash T$	0	1
0	0	1
1	1	0

$$Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n$$

$$Q_{n+1} = T \oplus Q_n$$

Lec 4

Aug 22, 2026

Flip Flop Summary:

1. Excitation Table

S-R FF		
S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	X

D-FF	
D	Q_{n+1}
0	0
1	1

J-K FF		
J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\overline{Q_n}$

T FF	
T	Q_{n+1}
0	Q_n
1	$\overline{Q_n}$

2. Excitation Requirement

$Q_n \rightarrow Q_{n+1}$	S	R
0 → 0	0	-
0 → 1	1	0
1 → 0	0	1
1 → 1	-	0

$Q_n \rightarrow Q_{n+1}$	D
0 → 0	0
0 → 1	1
1 → 0	0
1 → 1	1

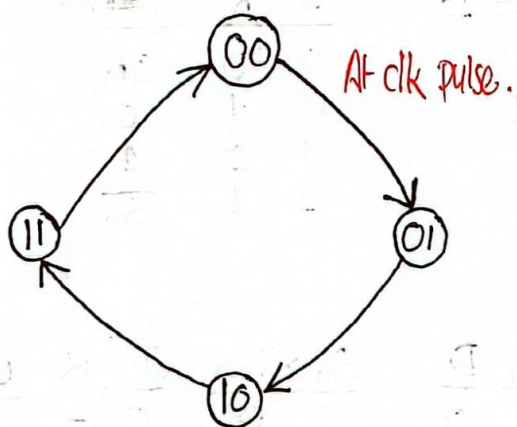
$Q_n \rightarrow Q_{n+1}$	J	K
0 → 0	0	-
0 → 1	1	-
1 → 0	-	1
1 → 1	-	0

$Q_n \rightarrow Q_{n+1}$	T
0 → 0	0
0 → 1	1
1 → 0	1
1 → 1	0

#1 2-bit Binary Counter: Use JK flip-flops.

◦ State Diagrams

• 4 states



◦ State Table:

PS	→	NS
00		01
01		10
10		11
11		00

#states = 4

$$\therefore \#FF = \lceil \log_2 4 \rceil = 2$$

◦ Transition Table:

• Represent states with variables

MSB	LSB	→	MSB	LSB
y_2	y_1		Y_2	Y_1
00			01	
01			10	
10			11	
11			00	

$y_2 \rightarrow Y_2$

$y_1 \rightarrow Y_1$

Excitation Tables

y_2	y_1	corresponds to y_2 FF		corresponds to y_1 FF	
		J_2	K_2	J_1	K_1
00		$0 \rightarrow 0$	0	0	1
01		$0 \rightarrow 1$	1	1	0
10		$1 \rightarrow 0$	0	0	1
11		$1 \rightarrow 1$	1	1	0

from excitation
requirement
table

K-Map:

J_2

y_2	y_1
0	1
1	0

$$J_2 = y_1$$

K_2

y_2	y_1
-	1
0	1

$$K_2 = y_1$$

J_1

y_2	y_1
1	0
1	1

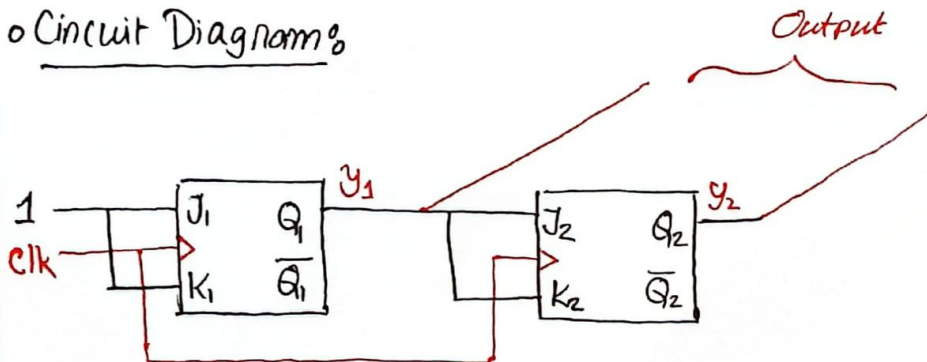
$$J_1 = 1$$

K_1

y_2	y_1
-	1
-	1

$$K_1 = 1$$

Circuit Diagram



#2 Sequence Detector : Overlapping

← Question 6 का शुरुआत

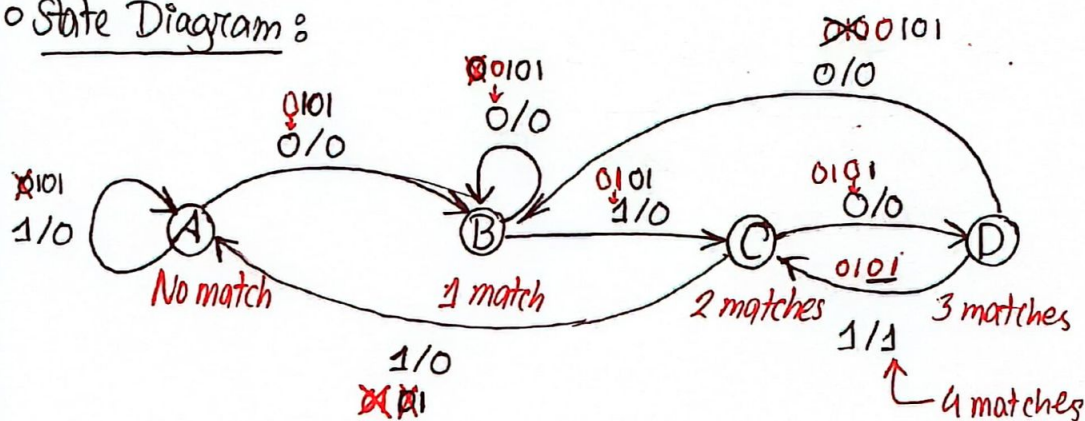
* $x \rightarrow 1011010101100011$

match $\rightarrow 0101$ $\rightarrow$ find

$z \rightarrow 000000001010000000$ ← Output

Use SR Flip-Flop.

◦ State Diagram :



◦ State Table :

NS, z

PS	x=0	x=1
A	B, 0	A, 0
B	B, 0	C, 0
C	D, 0	A, 0
D	B, 0	C, 1

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◦ Transition Table: Declare variables to store the states

• $\#FF = \#Variable = \log_2 4 = 2$
 $\hookrightarrow 4 \text{ states.}$

$y_2 y_1$	y_2, y_1, z	
	$x=0$	$x=1$
A = 00	01, 0	00, 0
B = 01	01, 0	10, 0
C = 10	11, 0	00, 0
D = 11	01, 0	10, 1

Assign values
to represent the state

◦ Excitation Table:

$y_2 y_1$	$S_2 R_2 (Y_2)$		$S_1 R_1 (Y_1)$	
	$x=0$	$x=1$	$x=0$	$x=1$
00	0-	0-	10	0-
01	0-	10	-0	01
10	-0	01	10	0-
11	01	-0	-0	01

Excitement requirement for SRFF

$Q_n \rightarrow Q_{n+1}$	S	R
0 → 0	0	-
0 → 1	1	0
1 → 0	0	1
1 → 1	-	0

Imp for
CT

• K-Map: 5 output variables. Each is a func of x, y_1, y_2

S_2 :

$y_2 y_1 \backslash x$	0	1
00	0	0
01	0	1
11	0	1
10	-	0

$S_2 = xy_1$

R_2 :

$y_2 y_1 \backslash x$	0	1
00	-	1
01	1	0
11	1	0
10	0	1

$R_2 = xy_1 + xy_1$

$R_2 = x \oplus y_1$

S_1 :

$y_2 y_1 \backslash x$	0	1
00	1	0
01	-	0
11	-	0
10	1	0

$S_1 = \bar{x}$

R_1 :

$y_2 y_1 \backslash x$	0	1
00	0	-
01	0	1
11	0	1
10	0	-

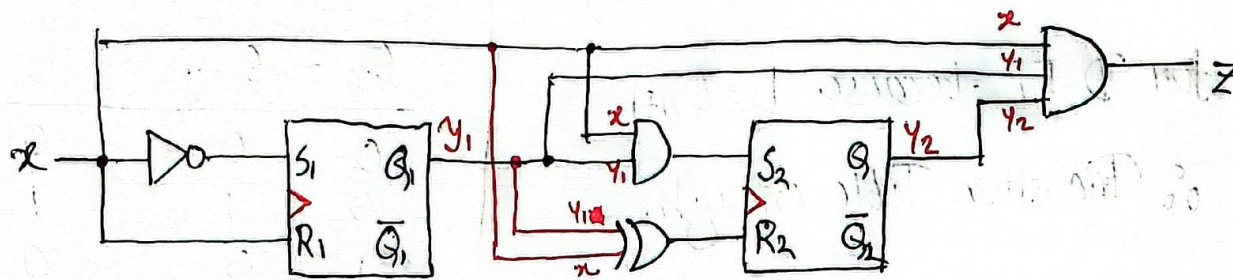
$R_1 = x$

Z :

$y_2 y_1 \backslash x$	0	1
00	0	0
01	0	0
11	0	1
10	0	0

$Z = xy_1 y_2$

Circuit Diagram:



x	Q_1	Q_2	z
0	0	0	0
0	1	0	0
1	0	1	1
1	1	0	0

x	Q_1	Q_2	z
0	0	0	0
0	1	0	0
1	0	1	1
1	1	0	0

#3 Sequence Detection with D-FF

⇒ Excitation Table not required

for D-FF because $D = Q_{n+1}$

∴ Transition Table enough.

K-Maps

From prev math Transition Table:

Excitation Table:

$Y_2 Y_1$	$D_2 (Y_2)$		$D_1 (Y_1)$	
	$x=0$	$x=1$	$x=0$	$x=1$
00	0	0	1	0
01	0	1	1	0
10	1	0	1	0
11	0	1	1	0

→ This table has the same information as the Transition Table
Thus this is redundant.

K-Map:

D_2

$Y_2 \backslash Y_1$	0	1
0	0	0
1	0	1
2	0	1
3	1	0

$$D_2 = \bar{x} \bar{Y}_1 Y_2 + x Y_1$$

D_1

$Y_2 \backslash Y_1$	0	1
0	1	0
1	1	0
2	1	0
3	1	0

$$D_1 = \bar{x}$$

$$Z = xy_1y_2 \quad [\text{from previous math}]$$

o Circuit Diagram:



L		Q
0	0	0
0	1	0
1	0	0
1	1	1
0	0	0
0	1	0
1	0	0
1	1	1

Lec 6

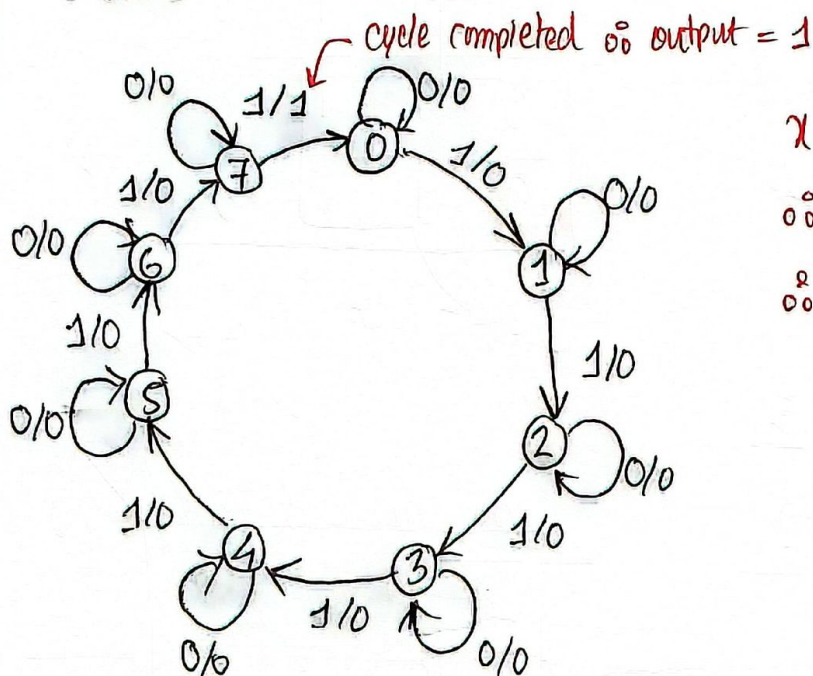
Aug 25, 2026

#1 Modulo N counter: using T FF

* $N=8$. Counts clock when $x=1$. $z = \begin{cases} 1, & \text{when count} = 0 \text{ (cycle completed)} \\ 0, & \text{when count} = 1..7 \end{cases}$

State Diagram:

• 8 states



$x=0$ means no count

$\therefore$ No state change

$\therefore$ Self loop.

State Table:

NS, Z

PS	$x=0$	$x=1$
S_0	$S_0, 0$	$S_1, 0$
S_1	$S_1, 0$	$S_2, 0$
S_2	$S_2, 0$	$S_3, 0$
S_3	$S_3, 0$	$S_4, 0$
S_4	$S_4, 0$	$S_5, 0$
S_5	$S_5, 0$	$S_6, 0$
S_6	$S_6, 0$	$S_7, 0$
S_7	$S_7, 0$	$S_0, 1$ ←

Transition Table:

$$\#FF = \lceil \log_2 8 \rceil = 3$$

Represent states with variables.

3 variables

	y_2	y_1	y_0	$\rightarrow$	Y_2	Y_1	Y_0	z
					$x=0$			$x=1$
S_0	0	0	0		000	0	001	0
S_1	0	0	1		001	0	010	0
S_2	0	1	0		010	0	011	0
S_3	0	1	1		011	0	100	0
S_4	1	0	0		100	0	101	0
S_5	1	0	1		101	0	110	0
S_6	1	1	0		110	0	111	0
S_7	1	1	1		111	0	000	1

Excitation Table:

	T_2, T_1, T_0		T_2, T_1, T_0	
	$x=0$		$x=1$	
y_2, y_1, y_0	$T_2(y_2)$	$T_1(y_1)$	$T_0(y_0)$	
	$x=0$	$x=1$	$x=0$	$x=1$
000	000	000	001	001
001	001	000	011	011
010	010	000	001	001
011	011	000	111	111
100	100	000	001	001
101	101	000	011	011
110	110	000	001	001
111	111	000	111	111

Excitation Requirement for TFF:

$Q_n \rightarrow Q_{n+1}$	T
0 → 0	0
0 → 1	1
1 → 0	1
1 → 1	0

for No change

no change

o K-Map: 4-kmaps $\rightarrow T_2, T_1, T_0, Z$; each of 4 variables: y_2, y_1, y_0, x

T_0 :

		y_0			
		00	01	11	10
y_2	00	0	1	1	0
	01	0	1	1	0
	11	0	1	1	0
	10	0	1	1	0

y_1

$x \rightarrow T_0 = x$

T_1 :

		y_0			
		00	01	11	10
y_2	00	0	0	1	0
	01	0	0	1	0
	11	0	0	1	0
	10	0	0	1	0

y_1

$x \rightarrow T_1 = xy_0$

T_2 :

		y_0			
		00	01	11	10
y_2	00	0	0	0	0
	01	0	0	1	0
	11	0	0	1	0
	10	0	0	0	0

y_1

$x \rightarrow T_2 = xy_0y_1$

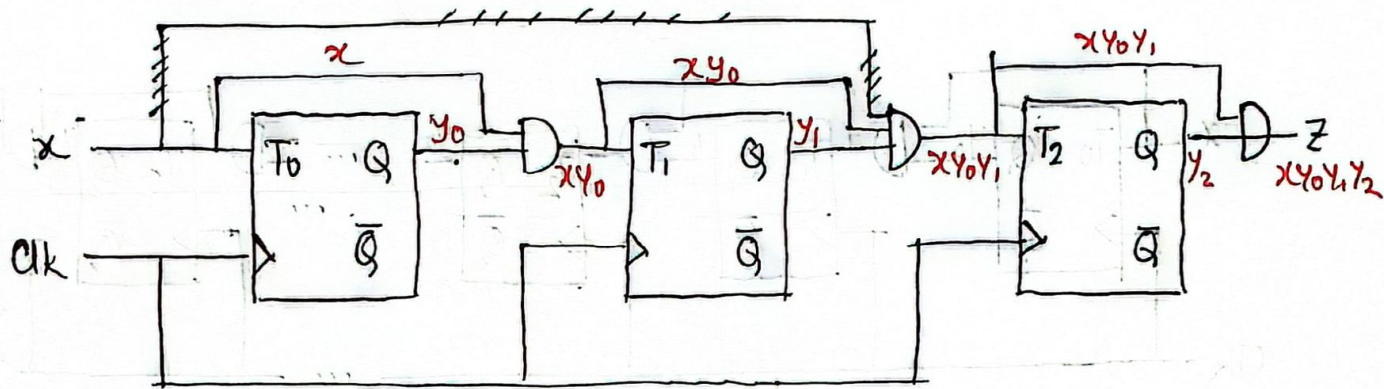
Z :

		y_0			
		00	01	11	10
y_2	00				
	01				
	11			1	
	10				

y_1

$x \rightarrow Z = xy_0y_1y_2$

o Circuit Diagram:



* $N = 128$

$\Rightarrow \#ff = \log_2 64 = 6 \quad \log_2 128 = 7$

$N = 2^7$ power 2^7 same power pattern.

oo $T_0 = x$

$T_1 = y_0 x$

$T_2 = y_1 y_0 x$

$T_3 = y_2 y_1 y_0 x$

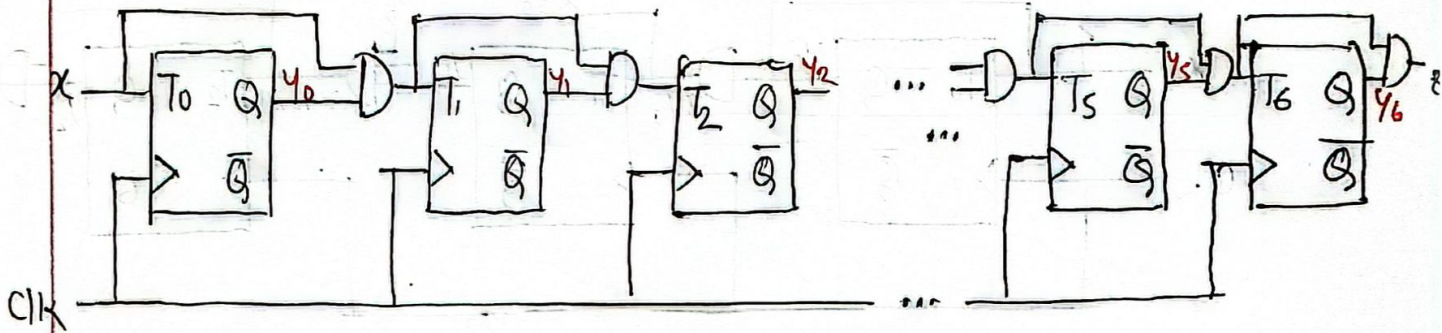
$T_4 = y_3 y_2 y_1 y_0 x$

$T_5 = y_4 y_3 y_2 y_1 y_0 x$

$T_6 = y_5 y_4 y_3 y_2 y_1 y_0 x$

$z = y_6 y_5 y_4 y_3 y_2 y_1 y_0 x$

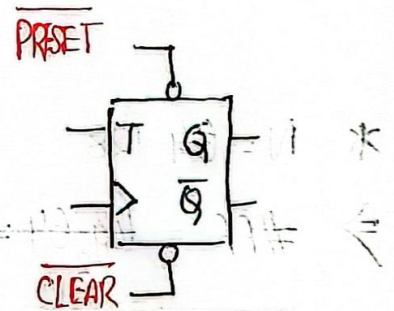
Circuit Diagram:



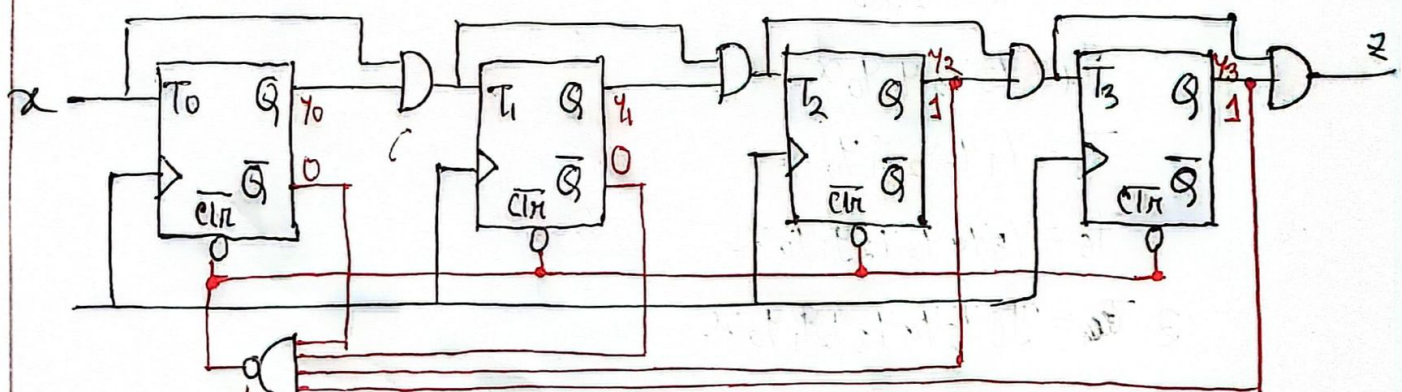
* $N=12$ ← not a power of 2.

⇒ Nearest upper 2's power $\rightarrow 16 \rightarrow 4$ FF

We need FFs with PRESET and CLEAR



CLEAR 0 0 (not clear) (FF reset)
 ↳ active low



NAND since Active low

Reset at 12 $\rightarrow 1100$

↓ ↓ ↓ ↓
 $Q_3 Q_2 Q_1 Q_0$

when state reaches 1100 (12)

clear is activated $\rightarrow$ Thus every FF becomes 0000

HW → Down Counter → PRESET needed.

Lec 7.

Aug 29, 26

Synchronous Sequential Circuit:

- Synthesis : Problem spec $\rightarrow$ Circuit
- Analysis : Circuit $\rightarrow$ What it does.

Steps:

1. State Diagram

2. State Table

→ 3. Transition # FF, state assignment

4. Transition Table

→ 5. Excitation Requirement

6. Excitation Table

7. K-Map

8. Circuit Diagram.

Analysis ↑

↓ Synthesis

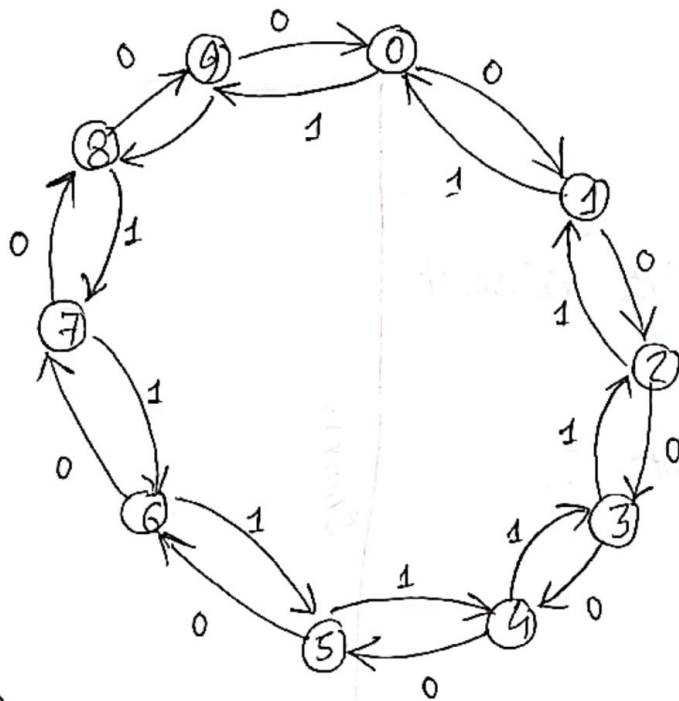
#1 Up-Down Counter: ~~Use~~ JK FF

• BCD counter / Decade counter (0-9)

• $x=0 \rightarrow$ Up
(Increment)

$x=1 \rightarrow$ Down
(Decrement)

• State Diagram:



~~State~~
• Transition Table:

$$\#FF = \lceil \log_2 10 \rceil = 4$$

$Q_n \rightarrow Q_{n+1}$	J	K
0 0	0	-
0 1	1	-
1 0	-	1
1 1	-	0

State Table:

Transition Table: #FF = $\lceil \log_2 10 \rceil = 4$

PS	NS		$Y_3 Y_2 Y_1 Y_0$	$Y_3 Y_2 Y_1 Y_0$	
	$x=0$	$x=1$		$x=0$	$x=1$
S_0	S_1	S_9	S_0 0000	0001	1001
S_1	S_2	S_0	S_1 0001	0010	0000
S_2	S_3	S_1	S_2 0010	0011	0001
S_3	S_4	S_2	S_3 0011	0100	0010
S_4	S_5	S_3	S_4 0100	0101	0011
S_5	S_6	S_4	S_5 0101	0110	0100
S_6	S_7	S_5	S_6 0110	0111	0101
S_7	S_8	S_6	S_7 0111	1000	0110
S_8	S_9	S_7	S_8 1000	1001	0111
S_9	S_0	S_8	S_9 1001	1011 0000	1000

Excitation Table:

$Y_3 Y_2 Y_1 Y_0$	$J_3 K_3$		$J_2 K_2$		$J_1 K_1$		$J_0 K_0$	
	$x=0$	$x=1$	$x=0$	$x=1$	$x=0$	$x=1$	$x=0$	$x=1$
0000	0-	1-	0-	0-	0-	0-	1-	1-
0001	0-	0-	0-	0-	1-	0-	-1	-1
0010	0-	0-	0-	0-	-0	-1	1-	1-
0011	0-	0-	1-	0-	-1	-0	-1	-1
0100	0-	0-	-0	-1	0-	1-	1-	1-
0101	0-	0-	-0	-0	1-	0-	-1	-1
0110	0-	0-	-0	-0	-0	-1	1-	1-
0111	1-	0-	-1	-0	-1	-0	-1	-1
1000	-0	-1	0-	1-	0-	1-	1-	1-
1001	-1	-0	0-	0-	0-	0-	-1	-1

o K-Maps: 8-Kaps, K-maps, each 5 variable (x, y_0, y_1, y_2, y_3)

$\overline{y_3} y_0$ $x=0$

$y_3 y_2$	$y_1 y_0$	00	01	11	10
00					
01					
11					
10					

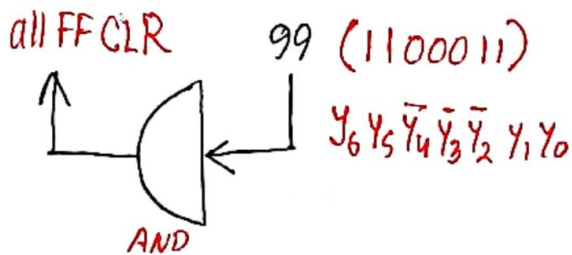
$y_3 y_2$ $x=1$

$y_1 y_0$	00	01	11	10
00				
01				
11				
10				

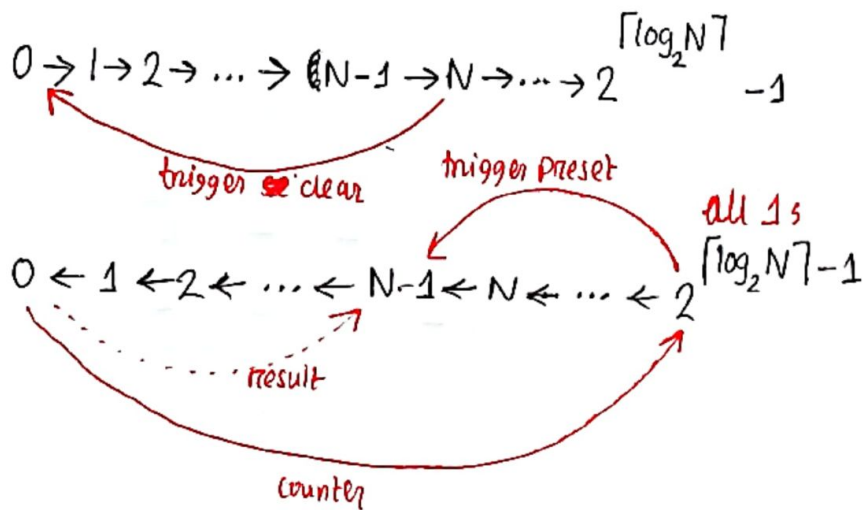
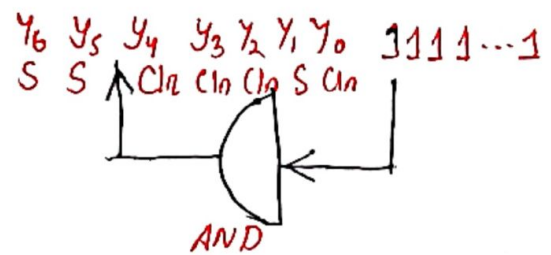
#2 Modulo-N Up-Down Counter (T-FF)

* $N=99$

• CLEAR when 99. (Up)



→ (1100010)
• 98 when all 1 (Down)



Lec 8

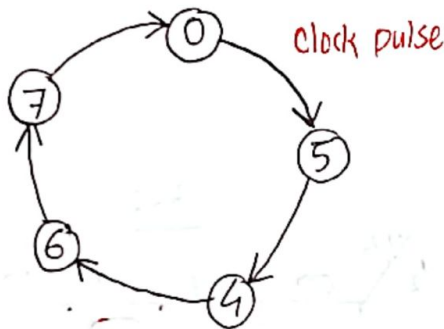
Aug 30, 2026

#1 Sequence Counter: No input, output Use JK FF

* $0 \rightarrow 5 \rightarrow 4 \rightarrow 6 \rightarrow 7$ (Given in question)

(In each clock pulse, the states of the FF must follow this sequence)

State Diagram:



State Tables:

PS	NS
S_0	S_5
S_5	S_4
S_4	S_6
S_6	S_7
S_7	S_0

Transition Tables:

	$Y_2 Y_1 Y_0$		$Y_2 Y_1 Y_0$
S_0	000		101 S_5
S_4	100		110 S_6
S_5	101		100 S_4
S_6	110		111 S_7
S_7	111		000 S_0
Rest			x (Don't care)

$$\#FF = \lceil \log_2 5 \rceil = 3$$

87?

Excitation Table:

$Y_2 Y_1 Y_0$	$\bar{Y}_2 K_2$	$J_1 K_1$	$\bar{J}_0 K_0$
000	1 -	0 -	1 -
100	- 0	1 -	0 -
101	- 0	0 -	- 1
110	- 0	- 0	1 -
111	- 1	- 1	- 1

Excitation Requirement for JKFF

$Q_n \rightarrow Q_{n+1}$	$\bar{J}$	K
0 0	0	-
0 1	1	-
1 0	-	1
1 1	-	0

K-Map:

J_2

$Y_2 \backslash Y_1 Y_0$	00	01	11	10
0	1	-	-	-
1	-	-	-	-

$$J_2 = 1$$

K_2

$Y_2 \backslash Y_1 Y_0$	00	01	11	10
0	-	-	-	-
1	0	0	1	0

$$K_2 = Y_1 Y_0$$

J_1

$Y_2 \backslash Y_1 Y_0$	00	01	11	10
0	0	-	-	-
1	1	0	-	-

$$J_1 = Y_2 \bar{Y}_0$$

K_1

$Y_2 \backslash Y_1 Y_0$	00	01	11	10
0	-	-	-	-
1	-	-	1	0

$$K_1 = Y_0$$

J_0

$Y_2 \backslash Y_1 Y_0$	00	01	11	10
0	1	-	-	-
1	0	-	-	1

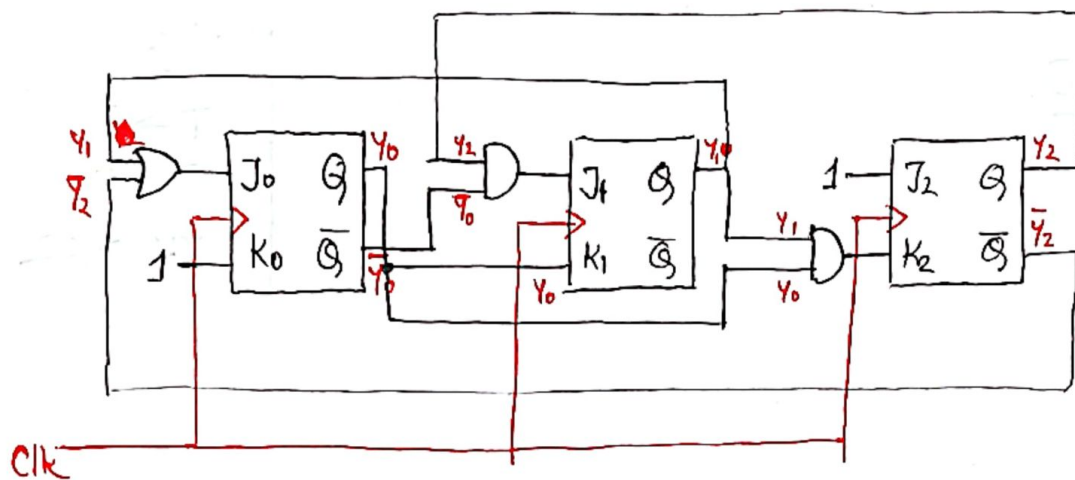
$$J_0 = \bar{Y}_2 + Y_1$$

K_0

$Y_2 \backslash Y_1 Y_0$	00	01	11	10
0	-	-	-	-
1	-	1	1	-

$$K_0 = 1$$

° Circuit Diagram :



⊛ Problem:

If the initial state of the FFs is not 0, 5, 4, 6, 7, the counter won't work properly.

Undefined state.

↳ it may loop through the undefined states.

Solution: (Synchronous)

Define all transitions from each undefined states.

So if starts with an undefined state we design the ckt so that it transitions to a defined state in the 1st pulse.

↳ New problem: 1st pulse is wasted

"Initialization problem"

Solution: (Asynchronous)

Asynchronous ~~RS~~ RESET/CLEAR.

CT, Final Conversion Between Flip-Flops: 5 marks question.

#1 T $\rightarrow$ D



Practice

T $\rightarrow$ D

D $\rightarrow$ T

JK $\rightarrow$ SR

Lec 9

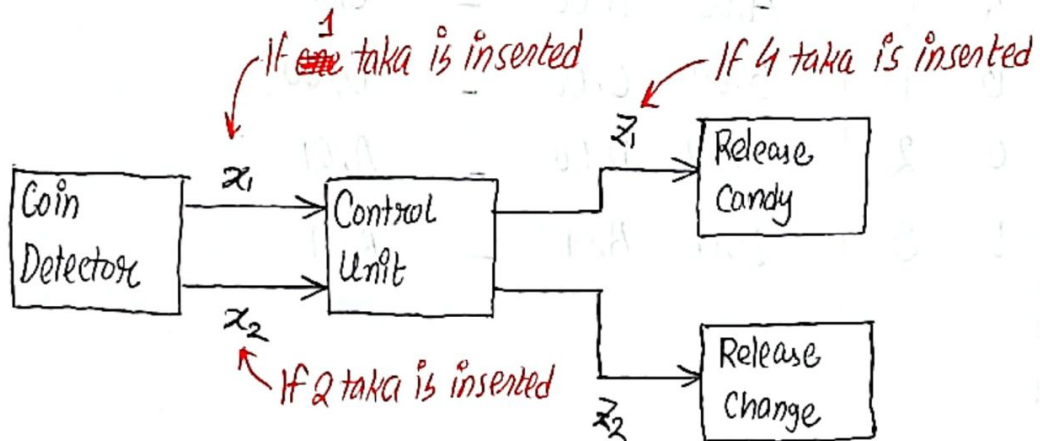
Sep 1, 2026

Vending Machine: Design the control unit.

- Only 1 type of product.

Similar:

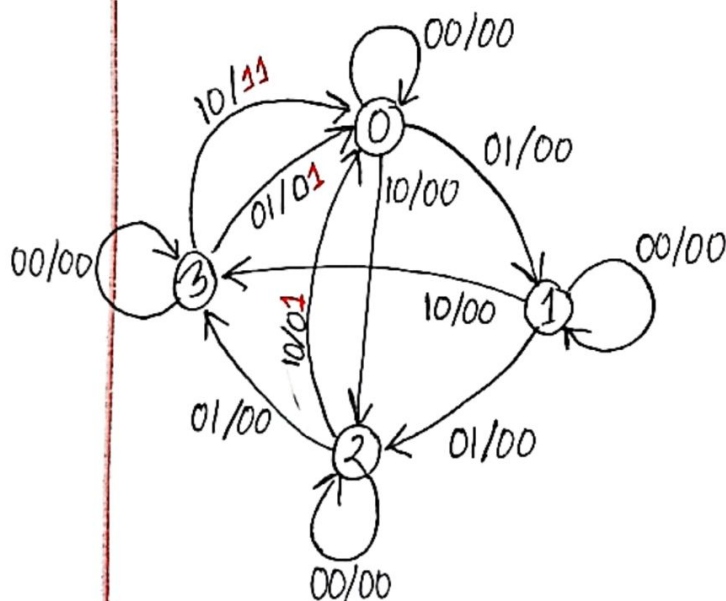
- Digital lock
- Traffic signal.



* 2 types of coins: 1Tk, 2Tk

Candy price: 4Tk

State Diagram:



Both input can't be 1 simultaneously
Because we can't insert 2 coins at
the same time.

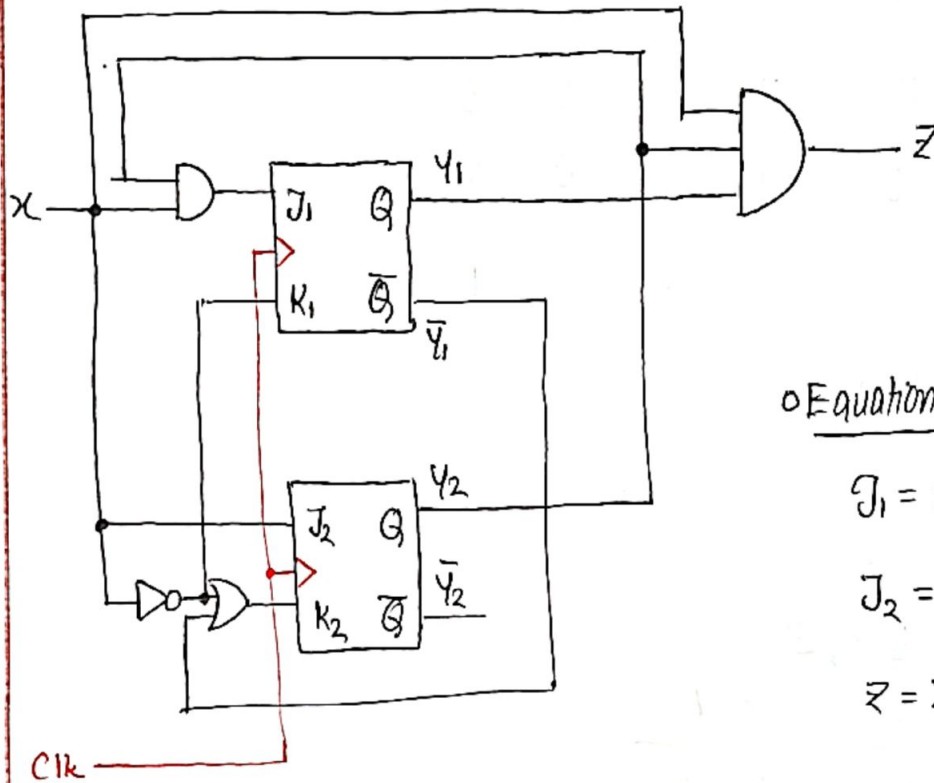
o State Tables

x_2x_1 PS		$N_5, Z_2 Z_1$			
		00	01	11	00
A	0	A,00	B,00	-	C,00
B	1	B,00	C,00	-	D,00
C	2	C,00	D,00	-	A,01
D	3	D,00	A,01	-	A,11



Sequential Circuit Analysis

#1



Equations:

$$J_1 = xy_2, \quad K_1 = \bar{x}$$

$$J_2 = x, \quad K_2 = \bar{x} + \bar{y}_1$$

$$z = xy_1y_2$$

K-Maps:

J₁:

y ₂ y ₁ \ x	0	1
00		
01		
11		1
10		1

J₂:

y ₂ y ₁ \ x	0	1
00		1
01		1
11		1
10		1

K₁:

y ₂ y ₁ \ x	0	1
00	1	
01	1	
11	1	
10	1	

K₂:

y ₂ y ₁ \ x	0	1
00	1	1
01	1	
11	1	
10	1	1

z:

y ₂ y ₁ \ x	0	1
00		
01		
11		1
10		

Excitation Table:

$Y_2 Y_1$	$J_2 K_2$		$J_1 K_1$		Z	
	$x=0$	$x=1$	$x=0$	$x=1$	$x=0$	$x=1$
00	01	11	01	00	0	0
01	01	10	01	00	0	0
11	01	10	01	10	0	1
10	01	11 11	01	10	0	0

Transition Table:

$Y_2 Y_1$	$Y_2 Y_1, Z$	
	$x=0$	$x=1$
00	00, 0	10, 0
01	00, 0	11, 0
11	00, 0	11, 1
10	00, 0	01, 0

State Table:

PS	NS, Z	
	$x=0$	$x=1$
$A \leftarrow 00$	A, 0	D, 0
$B \leftarrow 01$	A, 0	C, 0
$C \leftarrow 11$	A, 0	C, 1
$D \leftarrow 10$	A, 0	B, 0

o State Diagram :

